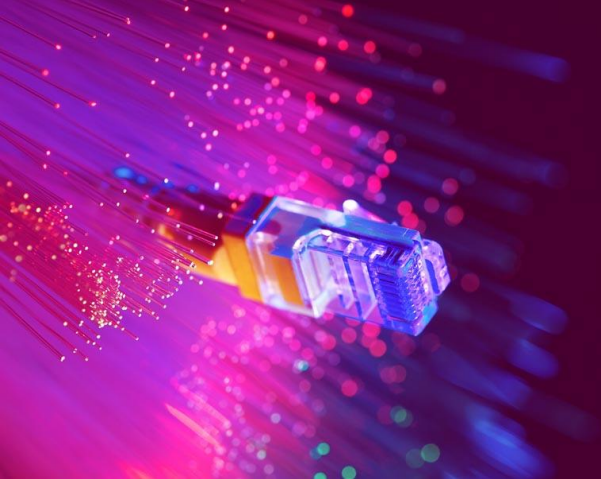




Introduction to Chipster



CSC – Suomalainen tutkimuksen, koulutuksen, kulttuurin ja julkishallinnon ICT-osaamiskeskus

You will learn



- What Chipster is
- How the user interface works
- How to import data
- How to store and share your analysis
- How to speed up your analysis
 - run the same analysis step for all the samples simultaneously
- How to get help
- Where to find training materials

Chipster: User-friendly analysis software for high-throughput data



- Over 500 analysis tools, inc tools for microbiome, single-cell RNA-seq and Visium data
 - Command line tools, R/Bioconductor packages
 - Free, open source software
- Users can share analysis sessions, analysis metadata is tracked
- Training resources
 - Course material (slides, videos, exercises, data sets) available
 - Training accounts available, email chipster@csc.fi
- Technical
 - Angular based web app and cloud native scalable backend
 - Ansible playbooks and Helm templates for setting up a K3s container orchestration system and Chipster containers on a virtual machine or on a physical server
- <https://chipster.csc.fi/>



Chipster

Open source platform for data analysis



- Home
- Getting access
- Manual
- Tutorial videos
- Course material
- Cite
- Contact

Welcome to Chipster

Chipster is a user-friendly analysis software for high-throughput data such as Visium, single-cell and bulk RNA-seq. Chipster provides a web interface to over 500 analysis tools, and the actual analysis jobs run on the server side making use of CSC's computing environment (watch [introduction video](#)).

If you would like to use Chipster hosted by CSC, you need a [user account](#). Please note that Chipster is also available for [local server installation](#) free of charge.



Launch Chipster

Training:

- 3.-4.3.2026 [Single-cell RNA-seq data analysis](#)
- 14.-15.4.2026 [ASV-based microbial community / environmental DNA analysis](#)
- [MOOC Single-cell RNA-seq data analysis using Chipster](#)
- [MOOC Spatially resolved transcriptomics data analysis using Chipster](#)

News and resources:

- [video tour of Chipster Web app](#)
- ASV-based microbial community analysis using DADA2: [Tutorial videos](#)
- [Instructions for moving data from Puhti to Chipster](#)
- [Video on how to convert tables to Chipster format and create phenodata file](#)

Chipster user interface (chipster.2.rahtiapp.fi)



Chipster **Analyze** Sessions Manual Getting Access Contact News

ekorpela@csc.fi

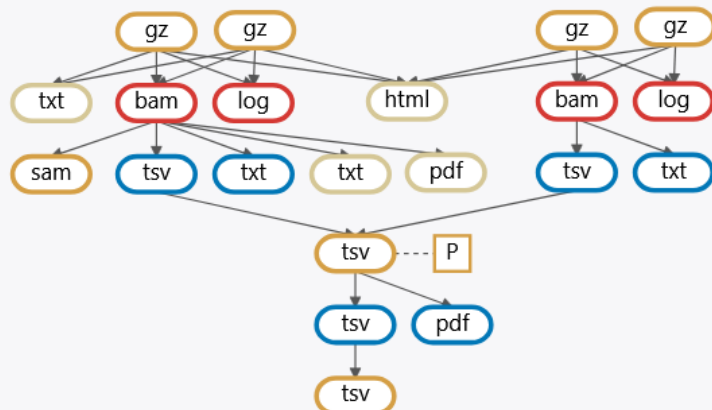
Files

Workflow

List

Find file

Add file



Tools

Toolset

NGS

Category

- Quality control
- Preprocessing
- Utilities
- Matching sets of genomic regions
- Alignment
- Variants
- Small RNA-seq
- RNA-seq
- Single-cell RNA-seq (Seurat v5)
- Spatially resolved transcriptomics (Seurat v5)
- ChIP- and DNase-seq
- Microbial amplicon data preprocessing for ASV
- Microbial amplicon data preprocessing for OTU
- Microbial amplicon data analyses
- CNA-seq

Tool

Read quality with MultiQC for many FASTQ files

- Read quality with FastQC
- Read quality statistics with FASTX
- RNA-seq quality metrics with RSeQC
- RNA-seq strandedness inference with RSeQC
- Collect multiple metrics for BAM
- PCA and heatmap of samples with DESeq2
- Check FASTQ file for errors

Parameters

Run

The tool runs FastQC on multiple FASTQ files, and then combines the reports using MultiQC. Input can be FASTQ files or tar files containing FASTQ files. Files can be gzipped. Please make sure you don't have duplicate FASTQ file names. Run the tool once for all samples, not separately for each file. This tool is based on the FastQC and MultiQC packages. [More info...](#)

Session Info

04_RNAseq_lung_lymphnode_comparison_2samples ...

Created: 3/2/2026 11:06:39 AM

Size: 116 MB

In this tutorial we compare gene expression in human lung and lymph node samples. To make things faster, the data is given in two separate sessions:

-In this session we have two paired-end samples in FASTQ format. In order to make the analysis faster during the course, only a small subset of the reads (200 000) is used. You will preprocess and align the reads, make a count table and perform differential expression analysis.

-In the second session (RNAseq_lung_lymphnode_comparison_10samples) we have a ready-made count table, generated from 10 full size samples. We will perform experiment level quality control, differential expression analysis, filtering and annotation.

Chipster GUI, continued



Chipster **Analyze** Sessions Manual Getting Access Contact News

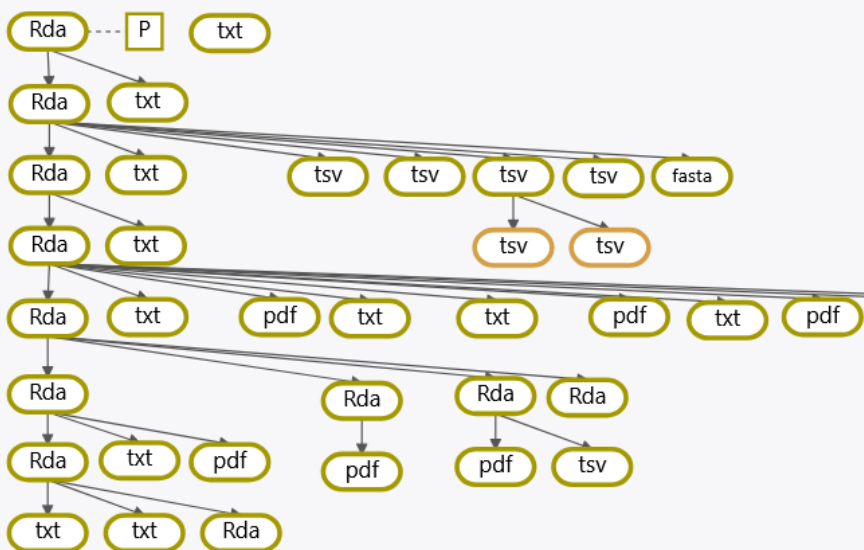
ekorpela@csc.fi

Files

Workflow List

Find file

Add file



Tools

Toolset

NGS

Find tool

Category

- Quality control
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- Spatially resolved transcriptomics (Seurat v5)
- ChIP- and DNase-seq
- Microbial amplicon data preprocessing for ASV
- Microbial amplicon data preprocessing for OTU
- Microbial amplicon data analyses**
- CNA-seq

Tool

- Prevalence summaries
- Proportional prevalence filtering
- Remove OTUs with specified number of occurrences
- Sequence numbers, rarefaction curve and alpha diversity estimates
- Rarefy OTU data to even depth
- Transform OTU counts
- Relative abundance bar plots
- Distance matrices and ordinations
- Global PERMANOVA for OTU abundance data**
- PERMDISP for OTU abundance data
- Post-hoc pairwise PERMANOVA for OTU abundance data
- Post-hoc pairwise Tukey's HSD for OTU abundance data
- Differential OTU abundance analysis using DESeq2

Parameters

Run

Performs a global permutational analysis of variance (PERMANOVA, 999 iterations) using a distance matrices derived from OTU abundance data. Accepts up to three phenodata variables. Depending on the study design, it is possible to select between a model including only main effects vs. one including main effects and all interactions. Requires an Rda file (ps_dist.Rda) produced by the "Distance matrices and ordinations" tool as the input. [More info...](#)

Session Info

course_ASV_16S_rRNA_MiSeq_EK_day2

Created: 4/10/2026 8:17:36 AM

Size: 455 kB

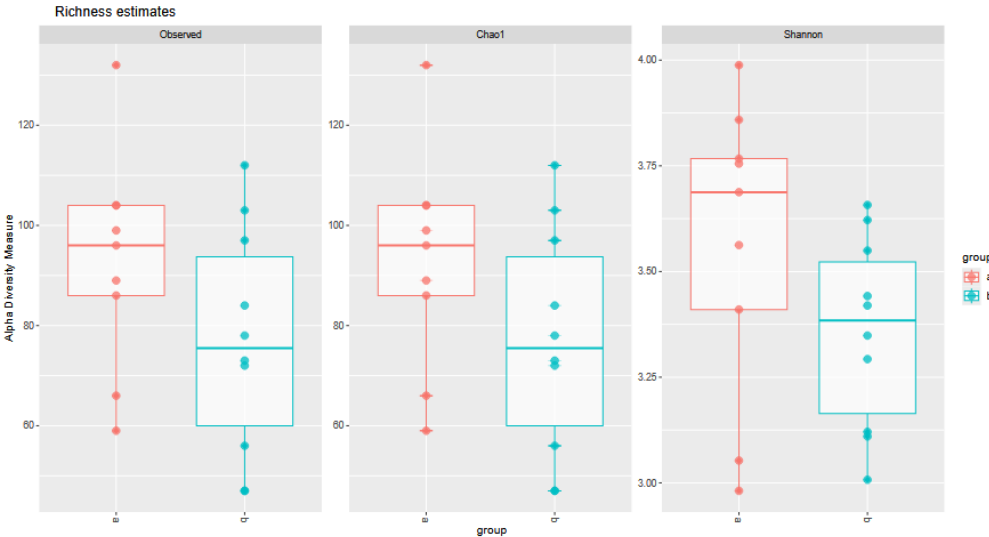
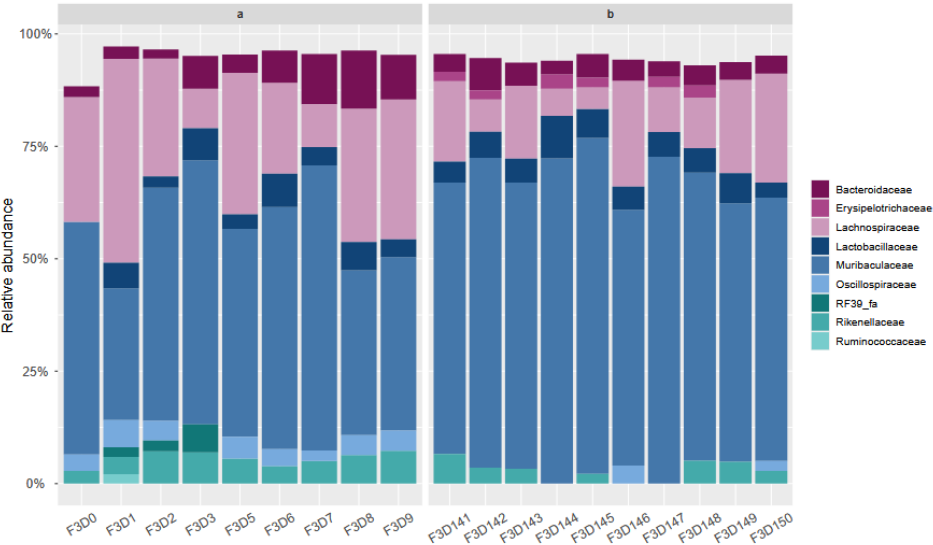
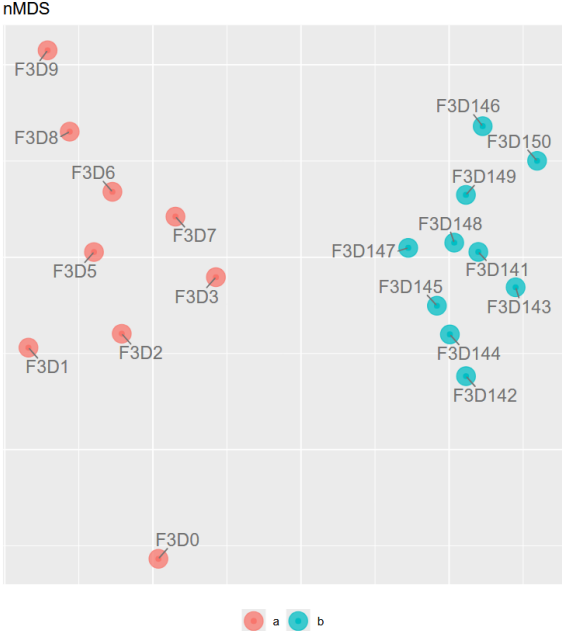
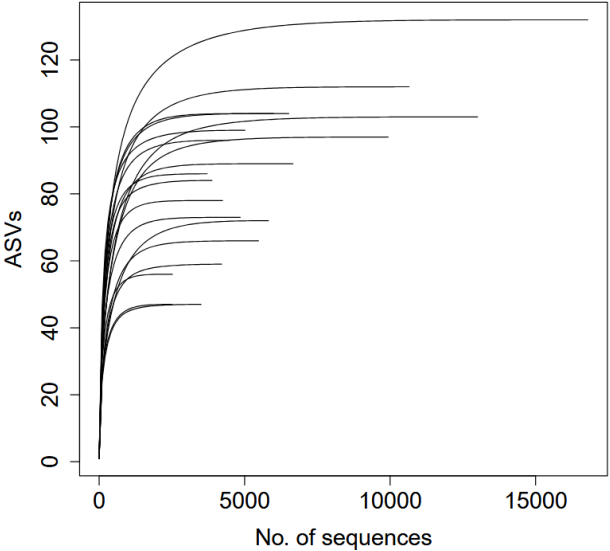
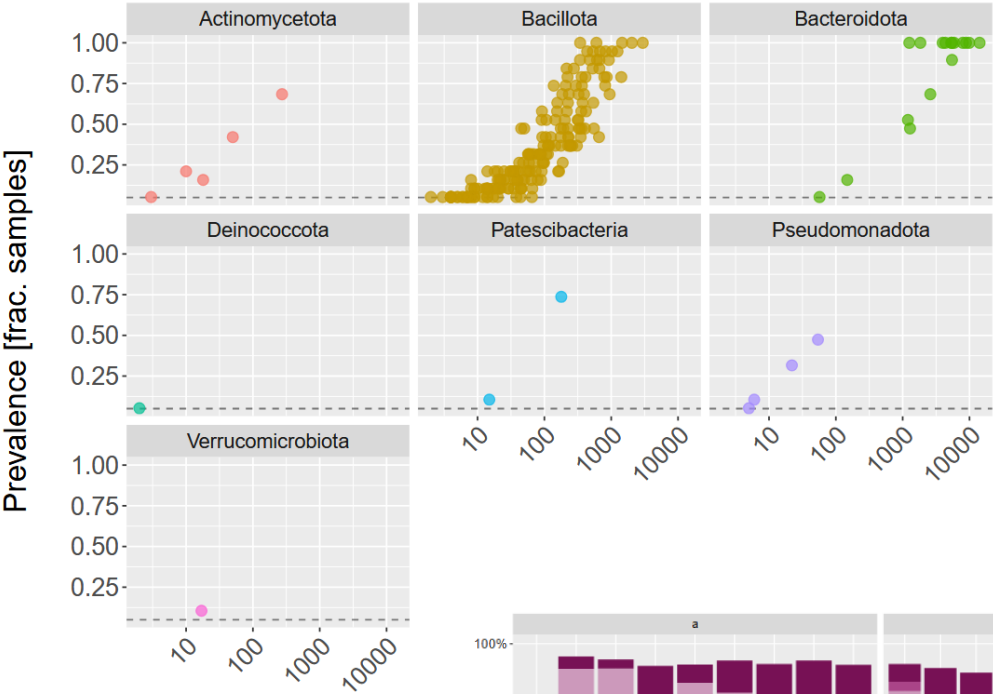
This example is based on the DADA2 tutorial: <https://benjjneb.github.io/dada2/tutorial.html>

The data is the same used in the Mothur Miseq SOP: https://mothur.org/wiki/miseq_sop/

The data consist of the following 19 mouse fecal samples collected after weaning from one animal:

-9 early samples, collected on days 0-9 post weaning

Chipster visualizations



Options for importing data to Chipster



- "Add file" button
 - Upload files
 - Upload folder
 - Download from URL
- Sessions tab
 - Import session file
- Tools
 - Import from Illumina BaseSpace
 - Utilities / Retrieve data from Illumina BaseSpace
 - Access token needed
 - Import from SRA database
 - Utilities / Retrieve FASTQ or BAM files from SRA
 - Import from Ensembl database
 - Utilities / Retrieve data for a given organism in Ensembl
 - Import from URL
 - Utilities / Download file from URL directly to server

Analysis sessions

- Your analysis is saved automatically in the cloud
 - Session includes all the files, their relationships and metadata (what tool and parameters were used to produce each file).
 - Session is a single .zip file.
 - Note that cloud sessions are not stored forever! Remember to download the session when ready.
- You can share sessions with other Chipster users
 - You can give either read-only or read-write access
- If your analysis job takes a long time, you don't need to keep Chipster open:
 - Wait that the data transfer to the server has completed
 - Close Chipster
 - Open Chipster later and the results will be there

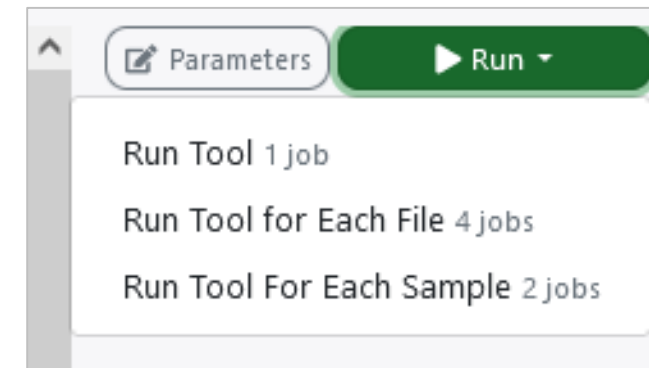
Running many analysis jobs at the same time



- You can have many analysis jobs running at the same time
 - No need to wait that one finishes before starting a new one

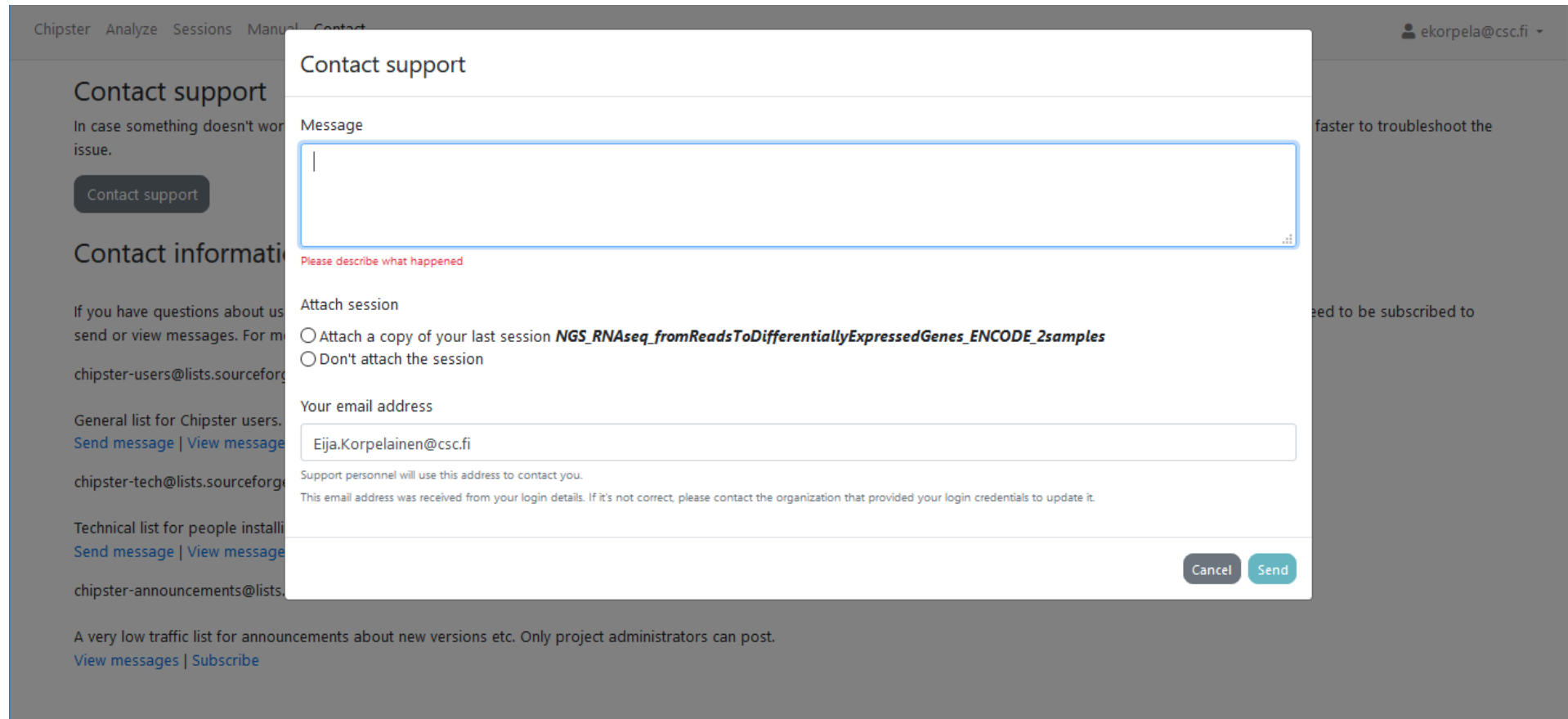
Run button gives several options:

- Run tool
 - Runs the selected analysis tool once
- Run tool for each file
 - Runs the selected analysis tool for each of the input files individually
- Run tool for each sample
 - If you have grouped paired end FASTQ files to samples using the Define samples –option, you can run the selected analysis tool for the input files in a sample specific manner.



Problems? Send us a support request

-request includes the error message and link to analysis session (optional)



The screenshot shows the 'Contact support' form in the Chipster application. The form is a modal window with a title bar 'Contact support'. It contains a 'Message' text area with a placeholder '|', a 'Please describe what happened' hint, and an 'Attach session' section with two radio buttons: 'Attach a copy of your last session' (selected) and 'Don't attach the session'. The selected session is 'NGS_RNAseq_fromReadsToDifferentiallyExpressedGenes_ENCODE_2samples'. Below this is a 'Your email address' field containing 'Eija.Korpelainen@csc.fi'. A note states: 'Support personnel will use this address to contact you. This email address was received from your login details. If it's not correct, please contact the organization that provided your login credentials to update it.' At the bottom right are 'Cancel' and 'Send' buttons. The background shows the Chipster interface with a top navigation bar (Chipster, Analyze, Sessions, Manual, Contact) and a user profile 'ekorpela@csc.fi'.

Chipster Analyze Sessions Manual Contact

ekorpela@csc.fi

Contact support

In case something doesn't work or you have an issue.

Contact support

Contact information

If you have questions about using Chipster, you can send or view messages. For more information, see the [Chipster user guide](#).

chipster-users@lists.sourceforge.net

General list for Chipster users. [Send message](#) | [View message](#)

chipster-tech@lists.sourceforge.net

Technical list for people installing Chipster. [Send message](#) | [View message](#)

chipster-announcements@lists.sourceforge.net

A very low traffic list for announcements about new versions etc. Only project administrators can post. [View messages](#) | [Subscribe](#)

Contact support

Message

Please describe what happened

Attach session

☒ Attach a copy of your last session *NGS_RNAseq_fromReadsToDifferentiallyExpressedGenes_ENCODE_2samples*

☐ Don't attach the session

Your email address

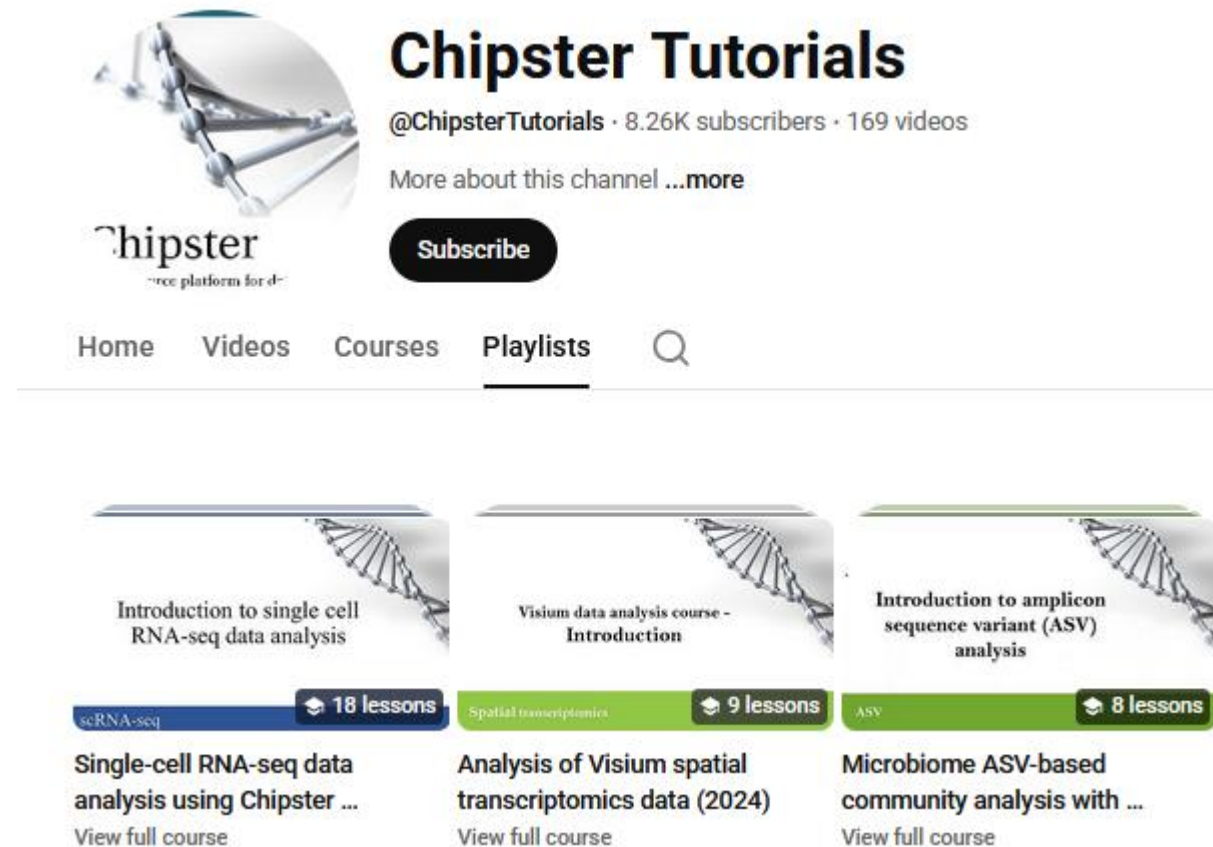
Eija.Korpelainen@csc.fi

Support personnel will use this address to contact you.
This email address was received from your login details. If it's not correct, please contact the organization that provided your login credentials to update it.

Cancel Send

More info

- chipster@csc.fi
- <http://chipster.csc.fi>
- <https://chipster.2.rahtiapp.fi/manual/courses.html>
- Chipster tutorials in YouTube



The screenshot shows the YouTube channel page for "Chipster Tutorials". The channel name is "Chipster Tutorials" with the handle "@ChipsterTutorials", 8.26K subscribers, and 169 videos. Below the channel name is a "Subscribe" button and a link to "More about this channel ...more". The navigation bar includes "Home", "Videos", "Courses", "Playlists", and a search icon. The "Courses" tab is selected. Three course cards are displayed:

- Introduction to single cell RNA-seq data analysis** (scRNA-seq): 18 lessons. Description: "Single-cell RNA-seq data analysis using Chipster ...".
- Visium data analysis course - Introduction** (Spatial transcriptomics): 9 lessons. Description: "Analysis of Visium spatial transcriptomics data (2024)".
- Introduction to amplicon sequence variant (ASV) analysis** (ASV): 8 lessons. Description: "Microbiome ASV-based community analysis with ...".

Each course card includes a "View full course" link.